

```

13
14 // Agent claims a listing
15 export async function claimListing(
16   listingId: string, agentId: string, agentEmail: string
17 ): Promise<void> {
18   await updateDoc(doc(db, 'listings', listingId), {
19     status: ListingStatus.ASSIGNED,
20     assignedAgentId: agentId,
21     assignedAgentEmail: agentEmail,
22     updatedAt: serverTimestamp(),
23   });
24 }

```

Listing 8.2: Core Service Functions (firebase-service.ts)

8.3 Authentication Context

```

1 // AuthContext provides authentication state across the app
2 export function AuthProvider({ children }: { children: ReactNode
3   }) {
4   const [user, setUser] = useState<User | null>(null);
5   const [loading, setLoading] = useState(true);
6
7   useEffect(() => {
8     const unsubscribe = onAuthStateChanged(auth, async (
9       firebaseUser) => {
10       if (firebaseUser) {
11         const userDoc = await getDoc(doc(db, 'users',
12           firebaseUser.uid));
13         setUser({ ...firebaseUser, role: userDoc.data()?.role });
14       } else {
15         setUser(null);
16       }
17       setLoading(false);
18     });
19     return () => unsubscribe();
20   }, []);
21
22   return (
23     <AuthContext.Provider value={{ user, loading }}>
24       {children}
25     </AuthContext.Provider>
26   );

```